CAMBRIDGE IGCSE™ & O LEVEL COMPUTER SCIENCE (0478 / 2210)

TOPIC 1.2: TEXT, SOUND AND IMAGES • DETAILED STUDY MATERIAL

Comprehensive Theory, Formulas, Worked Calculations & Cambridge Exam Notes

1.2.1 TEXT REPRESENTATION & CHARACTER SETS

1. What is a Character Set?

A **Character Set** is a defined list of all characters (letters, numbers, symbols, punctuation marks, and control codes) that are recognized and encoded into unique binary values by a computer system.
Whenever a key is pressed on a keyboard, the computer converts that character into its corresponding binary code.

2. ASCII vs Extended ASCII vs Unicode

Feature	Standard ASCII	Extended ASCII	Unicode (UTF-8 / UTF-16)
Bits per Character	7 bits	8 bits (1 byte)	16 bits or 32 bits (1 to 4 bytes)
Total Characters	128 characters (2^7)	256 characters (2^8)	Over 1,000,000 unique characters
Scope & Support	English letters (A-Z, a-z), numbers (0-9), punctuation, control codes.	Standard ASCII + non-English European letters and graphic symbols.	Universal standard: All world languages (Bengali, Arabic, Chinese, etc.), mathematical symbols, emojis.
Memory Requirement	Smallest file size	Small file size	Requires more storage space and memory per character.

3. Text File Size Calculation Formula

Text File Size (in Bytes) = Total Number of Characters × Number of Bytes per Character
• For 8-bit Extended ASCII: 1 character = 1 Byte
• For 16-bit Unicode: 1 character = 2 Bytes
• To convert Bytes to Kibibytes (KiB): Divide total Bytes by 1024.

1.2.2 SOUND REPRESENTATION & SAMPLING

1. Analogue to Digital Conversion (ADC)

• Sound in the real world is **analogue** (continuous sound waves).
• Computers cannot store continuous waves; they only process discrete binary numbers (0s and 1s).
• An **Analogue-to-Digital Converter (ADC)** converts analogue audio into digital format by capturing sound **samples** at regular time intervals.
• When playing sound back through speakers, a **Digital-to-Analogue Converter (DAC)** converts the binary data back into analogue audio waves.

2. Key Sound Terms & Definitions

• **Sample Rate:** The number of sound samples captured per second. It is measured in **Hertz (Hz)** or **Kilohertz (kHz)**. Example: CD quality audio is 44,100 Hz (44.1 kHz).
• **Sample Resolution (Bit Depth):** The number of bits used to store the amplitude/volume value of each sound sample. Example: 16-bit or 24-bit audio.
• **Audio Channels:** Mono (1 channel) or Stereo (2 channels - left and right).

3. The Quality vs File Size Trade-Off

Change Made	Effect on Sound Quality	Effect on File Size & Bandwidth
Higher Sample Rate (e.g. 44.1 kHz → 96 kHz)	Sound is much closer to original analogue wave; higher pitch accuracy.	File size increases significantly; requires more download bandwidth.
Higher Bit Depth (e.g. 8-bit → 16-bit / 24-bit)	Larger dynamic range; quieter audio details and nuances captured clearly.	File size increases proportionally.

4. Sound File Size Calculation Formula

Sound File Size (bits) = Sample Rate (Hz) × Sample Resolution (bits) × Length (seconds) × Number of Channels
• Step 1: Calculate total bits.
• Step 2: Convert to Bytes by dividing by 8.
• Step 3: Convert Bytes to Kilobytes (KB) (÷ 1000) or Kibibytes (KiB) (÷ 1024), and Megabytes (MB) (÷ 1,000,000) or Mebibytes (MiB) (÷ 1,048,576).

1.2.3 IMAGE REPRESENTATION & BITMAPS

1. How Bitmap Images are Formed

• A **Bitmap image** is composed of a two-dimensional grid of tiny individual squares called **pixels (picture elements)**.
• Each pixel has a specific binary value that represents its colour.
• **Image Resolution:** The total dimensions of the image measured as Width in pixels × Height in pixels (e.g. 1920 × 1080 = 2,073,600 total pixels).
• **Colour Depth (Bit Depth):** The number of bits allocated to represent the colour of each individual pixel.
  • 1-bit colour depth = 2^1 = 2 colours (Monochrome: Black and White).
  • 8-bit colour depth = 2^8 = 256 colours.
  • 24-bit colour depth (True Colour) = 2^24 = 16,777,216 unique colours (8 bits Red, 8 bits Green, 8 bits Blue).

2. Image File Size Calculation Formula

Image File Size (bits) = Image Width (pixels) × Image Height (pixels) × Colour Depth (bits)
• Step 1: Total bits = Width × Height × Colour Depth.
• Step 2: Total Bytes = Total bits ÷ 8.
• Step 3: To convert to Mebibytes (MiB), divide Bytes by 1,048,576 (or divide by 1024 twice).

3. Image Metadata

Metadata is extra information about the image stored inside the file header alongside the raw pixel data. Examples of metadata include: Image dimensions (width & height), colour depth, date and time of creation,

camera model, shutter speed/ISO, GPS location, and file format.

1.2.4 DATA COMPRESSION (LOSSY, LOSSLESS & RLE)

1. Why is Data Compression Needed?

1. **Reduces file size:** Saves valuable storage capacity on hard drives, SSDs, and cloud servers.
2. **Faster transmission & streaming:** Reduces upload, download, and web-page loading times.
3. **Saves bandwidth:** Consumes less network data when sending files over the Internet.

2. Lossy vs Lossless Compression Comparison

Feature	Lossy Compression	Lossless Compression
How it Works	Permanently removes unnecessary, redundant, or imperceptible data (e.g. sounds outside human hearing range, subtle colour differences humans cannot see).	Reduces file size by identifying repeated patterns and encoding them efficiently without removing any data.
Original File Reconstruction	The original file **cannot** be recreated exactly. Some quality is permanently lost.	The original file is **100% restored** bit-for-bit with zero loss in quality.
Compression Ratio	Very high file size reduction (up to 90% smaller).	Moderate file size reduction.
Suitable File Types	Photographs (**JPEG**), Audio/Music (**MP3**), Video (**MP4**).	Text documents (**PDF, DOCX**), Program code / Executables (**EXE, ZIP**), High-detail graphics (**PNG, GIF, FLAC**).

3. Run-Length Encoding (RLE)

• **Run-Length Encoding (RLE)** is a form of **lossless** data compression.
• It finds consecutive sequences of repeated identical data items (called "runs") and stores them as a single data value and its frequency count: `[count, value]`.
• **Example (Text/Colour):**
Uncompressed: `W W W W W  B B B  W W  G G G G` (14 data items)
RLE Compressed: `5W 3B 2W 4G` (or `05 W 03 B 02 W 04 G` — only 8 items, saving 43% space).